

torch.autograd.functional.vjp#

<https://pytorch.org/docs/stable/generated/torch.autograd.functional.vjp.html>

Description:

Compute the dot product between a vector v and the Jacobian of the given function at the point given by the inputs. `func` (function) – a Python function that takes Tensor inputs and returns a tuple of Tensors or a Tensor. `inputs` (tuple of Tensors or Tensor) – inputs to the function `func`. `v` (tuple of Tensors or Tensor) – The vector for which the vector Jacobian product is computed. Must be the same size as the output of `func`. This argument is optional when the output of `func` contains a single element and (if it is not provided) will be set as a Tensor containing a single 1. `create_graph` (bool, optional) – If True, both the output and result will be computed in a differentiable way. Note that when `strict` is False, the result can not require gradients or be disconnected from the inputs. Defaults to False.

Function Signature

```
torch.autograd.functional.vjp(func, inputs, v=None,  
create_graph=False, strict=False)[source]#
```

Parameters

Returns

tuple with:

Return type

Returns:

output (tuple)

Code Examples

```
>>> def exp_reducer(x):
...     return x.exp().sum(dim=1)
>>> inputs = torch.rand(4, 4)
>>> v = torch.ones(4)
>>> vjp(exp_reducer, inputs, v)
(tensor([5.7817, 7.2458, 5.7830, 6.7782]),
 tensor([[1.4458, 1.3962, 1.3042, 1.6354],
         [2.1288, 1.0652, 1.5483, 2.5035],
         [2.2046, 1.1292, 1.1432, 1.3059],
         [1.3225, 1.6652, 1.7753, 2.0152]]))
```

```
>>> vjp(exp_reducer, inputs, v, create_graph=True)
(tensor([5.7817, 7.2458, 5.7830, 6.7782], grad_fn=
<SumBackward1>),
 tensor([[1.4458, 1.3962, 1.3042, 1.6354],
         [2.1288, 1.0652, 1.5483, 2.5035],
         [2.2046, 1.1292, 1.1432, 1.3059],
         [1.3225, 1.6652, 1.7753, 2.0152]], grad_fn=
<MulBackward0>))
```

```
>>> def adder(x, y):
...     return 2 * x + 3 * y
>>> inputs = (torch.rand(2), torch.rand(2))
>>> v = torch.ones(2)
>>> vjp(adder, inputs, v)
(tensor([2.4225, 2.3340]),
 (tensor([2., 2.]), tensor([3., 3.])))
```

Detailed Documentation

Documentation

Compute the dot product between a vector v and the Jacobian of the given function at the point given by the inputs.

`func` (function) – a Python function that takes Tensor inputs and returns a tuple of Tensors or a Tensor.

`inputs` (tuple of Tensors or Tensor) – inputs to the function `func`.

`v` (tuple of Tensors or Tensor) – The vector for which the vector Jacobian product is computed. Must be the same size as the output of `func`. This argument is optional when the output of `func` contains a single element and (if it is not provided) will be set as a Tensor containing a single 1.

`create_graph` (bool, optional) – If True, both the output and result will be computed in a differentiable way. Note that when `strict` is False, the result can not require gradients or be disconnected from the inputs. Defaults to False.

`strict` (bool, optional) – If True, an error will be raised when we detect that there exists an input such that all the outputs are independent of it. If False, we return a Tensor of zeros as the vjp for said inputs, which is the expected mathematical value. Defaults to False.

`func_output` (tuple of Tensors or Tensor): output of `func(inputs)`

`vjp` (tuple of Tensors or Tensor): result of the dot product with the same shape as the inputs.

